Efficient and Robust Automated Machine Learning

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Abstract

The success of machine learning in a broad range of applications has led to an ever-growing demand for machine learning systems that can be used off the shelf by non-experts. To be effective in practice, such systems need to automatically choose a good algorithm and feature preprocessing steps for a new dataset at hand, and also set their respective hyperparameters. Recent work has started to tackle this *automated machine learning (AutoML)* problem with the help of efficient Bayesian optimization methods. Building on this, we introduce a robust new AutoML system based on scikit-learn (using 15 classifiers, 14 feature preprocessing methods, and 4 data preprocessing methods, giving rise to a structured hypothesis space with 110 hyperparameters). This system, which we dub AUTO-SKLEARN, improves on existing AutoML methods by automatically taking into account past performance on similar datasets, and by constructing ensembles from the models evaluated during the optimization. Our system won the first phase of the ongoing ChaLearn AutoML challenge, and our comprehensive analysis on over 100 diverse datasets shows that it substantially outperforms the previous state of the art in AutoML. We also demonstrate the performance gains due to each of our contributions and derive insights into the effectiveness of the individual components of AUTO-SKLEARN.

1 Introduction

Machine learning has recently made great strides in many application areas, fueling a growing demand for machine learning systems that can be used effectively by novices in machine learning. Correspondingly, a growing number of commercial enterprises aim to satisfy this demand (*e.g.*, BigML.com, Wise.io, SkyTree.com, RapidMiner.com, Dato.com, Prediction.io, DataRobot.com, Microsoft's Azure Machine Learning, Google's Prediction API, and Amazon Machine Learning). At its core, every effective machine learning service needs to solve the fundamental problems of deciding which machine learning algorithm to use on a given dataset, whether and how to preprocess its features, and how to set all hyperparameters. This is the problem we address in this work.

More specifically, we investigate automated machine learning (AutoML), the problem of automatically (without human input) producing test set predictions for a new dataset within a fixed computational budget. Formally, this AutoML problem can be stated as follows:

Definition 1 (AutoML problem)**.** *For $i = 1, \ldots, n+m$, let $\boldsymbol{x}_i \in \mathbb{R}^d$ denote a feature vector and $y_i \in Y$ the corresponding target value. Given a training dataset $D_{train} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$ and the feature vectors $\boldsymbol{x}_{n+1}, \ldots, \boldsymbol{x}_{n+m}$ of a test dataset $D_{test} = \{(\boldsymbol{x}_{n+1}, y_{n+1}), \ldots, (\boldsymbol{x}_{n+m}, y_{n+m})\}$ drawn from the same underlying data distribution, as well as a resource budget b and a loss metric $\mathcal{L}(\cdot, \cdot)$, the AutoML problem is to (automatically) produce test set predictions $\hat{y}_{n+1}, \ldots, \hat{y}_{n+m}$. The loss of a solution $\hat{y}_{n+1}, \ldots, \hat{y}_{n+m}$ to the AutoML problem is given by $\frac{1}{m} \sum_{j=1}^{m} \mathcal{L}(\hat{y}_{n+j}, y_{n+j})$.*